

	$= 150 + 2.0 \sin 75^\circ = 205.0 \text{ N/m}$
6	Answer: C Since the balloon is in equilibrium, upthrust on balloon = weight of balloon and helium + force by spring $\rho_{air} V_{balloon} g = \rho_{helium} V_{balloon} g + m_{balloon} g + kx$ $(1.29)(5.0)g = (0.180)(5.0)g + \left(\frac{3.5}{1000}\right)g + (100)x$ $x = 0.544 \text{ m}$
7	Answer: C